EXPT NO: 5(A)	A PYTHON PROGRAM TO SOLVE A LPP FOR CONSTRAINT OPTIMIZATION
DATE:	

AIM

To solve a Linear Programming Problem (LPP) for constraint optimization using Python.

PROCEDURE:

1.INSTALL REQUIRED LIBRARIES:

- Make sure you have numpy and scipy installed. You can install them using pip if n already installed.

  -bash

  -Copy the code

  -pip install numpy scipy

2.DEFINE THE PROBLEM:

- Identify the objective function to maximize or minimize.
- Determine the constraints (equality and inequality).

3.SET UP THE PROBLEM IN PYTHON:

- Use the scipy.optimize.linprog function to set up and solve the LPP.

4.RUN THE CODE:

- Execute the Python code to get the optimized result.

5.ANALYZE THE OUTPUT:

- Interpret the results and validate the solution against the problem constraints.

PROBLEM - 1: SOLVE THE FOLLOWING LINEAR PROGRAMMING PROBLEM

Maximize $z = 3x1 + 2x2$ Subject to Constraints:

- $2x1 + x2 \leq 20$,
- $4x1 - 5x2 \leq 10$,
- $x1 \geq 0, x2 \geq 0$.

CODE

```
from scipy.optimize import linprog
c = [-3, -2]
A = [
 [2, 1], [4, -5]
 ]
b = [20, 10]
 x_bounds = (0, None)
 y_bounds = (0, None)
 result = linprog(c, A_ub=A, b_ub=b, bounds=[x_bounds, y_bounds], method='highs')
 print("Optimal value:", -result.fun)
print("Values of x:", result.x)
```

OUTPUT

Optimal value: 40.0

Values of x: [0. 20.]

PROBLEM - 2: SOLVE THE FOLLOWING LINEAR PROGRAMMING PROBLEM

Maximize: $Z=5x+4y$ Subject to:

$x+y\leq 10$,

$2x+y\leq 15$,

$x,y\geq 0$

CODE

```
from scipy.optimize import linprog
c = [-5, -4]
A = [
[1, 1], [2, 1]
]
 b = [10, 15]
 x_bounds = (0, None)
```

```
y_bounds = (0, None)
 result = linprog(c, A_ub=A, b_ub=b, bounds=[x_bounds, y_bounds], method='highs')
 print("Optimal value:", -result.fun)
print("Values of x:", result.x)
```

OUTPUT

```
Optimal value: 50.0
Values of x: [5. 5.]
```

RESULT

Thus, the Linear Programming Problem (LPP) by Constraint Optimization using Python program has been implemented successfully.

EXPT NO: 5(B)	A PYTHON PROGRAM TO SOLVE A LPP BY GRAPHICAL METHOD
DATE:	

AIM:

To Solve a Linear programming problem by graphical method using Python.

PROCEDURE:

1. INSTALL REQUIRED LIBRARIES:

- Make sure that the librariesnumpy,scipy and matplotlib are installed.
- They can be installed using the following command:

!pip install numpyscipy, matplotlib

2. DEFINE THE PROBLEM:

- Identify the objective function to maximize or minimize.
- Determine the constraints (equality and inequality).

3. SET UP THE PROBLEM IN PYTHON:

- Use the scipy.optimize.linprog function to set up and solve the LPP.

4. RUN THE CODE:

- Execute the Python code to get the optimized result.

5. ANALYZE THE OUTPUT:

- Interpret the results and validate the solution against the problem constraints.

PROBLEM – 1:

Solve the following Linear Programming Problem Minimize z = 50x1 + 100x2 Subject to Constraints:

- 7x1 + 2x2 ≥ 28,
- 3x1 + 12x2 ≥ 24
- x1 ≥ 0, x2 ≥ 0

CODE

```
import numpy as np
```

```
import matplotlib.pyplot as plt
from scipy.optimize import linprog
c = [-4, -3]
A = [
    [1, 2],
    [3, 1]
]
b = [8, 9]
x_bounds = (0, None)
y_bounds = (0, None)
result = linprog(c, A_ub=A, b_ub=b,
                 bounds=[x_bounds, y_bounds],
                 method='highs')
x_opt, y_opt = result.x
z_opt = -result.fun
x_vals = np.linspace(0, 10, 400)
y1 = (8 - x_vals) / 2
y2 = (9 - 3 * x_vals)
plt.figure(figsize=(8, 8))
plt.plot(x_vals, y1, label=r'$x + 2y \leq 8$')
plt.plot(x_vals, y2, label=r'$3x + y \leq 9$')
y_feasible = np.minimum(y1, y2)
y_feasible = np.maximum(y_feasible, 0)
plt.fill_between(x_vals, 0, y_feasible, where=(y_feasible >= 0), alpha=0.5)
plt.plot(x_opt, y_opt, 'ro',
         label=f'Optimal Solution (x={x_opt:.2f}, y={y_opt:.2f})')
plt.xlim(0, 10)
```

```
plt.ylim(0, 10)

plt.xlabel('x')

plt.ylabel('y')

plt.title('Graphical Solution of Linear Programming Problem')

plt.legend()

plt.grid(True)

plt.show()

print("Optimal Solution : x =", round(x_opt,2),

    ", y =", round(y_opt,2),

    ", z =", round(z_opt,2))
```

OUTPUT

Optimal Solution : x = 3.6, y = 1.4, z = 320.0

PROBLEM - 2:

Maximize:

$$Z=5x+4yZ=5x+4y$$

Subject to:

$x+y\leq 6x+y\leq 6x+3y\leq 9x+3y\leq 9x\geq 0,\ \ y\geq 0x\geq 0,y\geq 0$

CODE

```
import numpy as np

import matplotlib.pyplot as plt

from scipy.optimize import linprog

c = [-5, -4]

A = [

   [1, 1],

   [1, 3]

]

b = [6, 9]

x_bounds = (0, None)
```

```
y_bounds = (0, None)
result = linprog(c, A_ub=A, b_ub=b,
                 bounds=[x_bounds, y_bounds],
                 method='highs')
x_opt, y_opt = result.x
z_opt = -result.fun
x_vals = np.linspace(0, 10, 400)
y1 = (6 - x_vals)
y2 = (9 - x_vals) / 3
plt.figure(figsize=(8, 8))
plt.plot(x_vals, y1, label=r'$x + y \leq 6$')
plt.plot(x_vals, y2, label=r'$x + 3y \leq 9$')
y_feasible = np.minimum(y1, y2)
y_feasible = np.maximum(y_feasible, 0)
plt.fill_between(x_vals, 0, y_feasible, where=(y_feasible >= 0), alpha=0.5)
plt.plot(x_opt, y_opt, 'ro',
         label=f'Optimal Solution (x={x_opt:.2f}, y={y_opt:.2f})')
plt.xlim(0, 10)
plt.ylim(0, 10)
plt.xlabel('x')
plt.ylabel('y')
plt.title('Graphical Solution of Linear Programming Problem')
plt.legend()
plt.grid(True)
plt.show()
print("Optimal Solution : x =", round(x_opt,2),
      ", y =", round(y_opt,2),
```

```
", z =", round(z_opt,2))
```

OUTPUT

Optimal Solution : x = 4.5 , y = 1.5 , z = 30.0

RESULT

Thus, the Linear Programming Problem (LPP) by Graphical method using Python program has been implemented successfully.